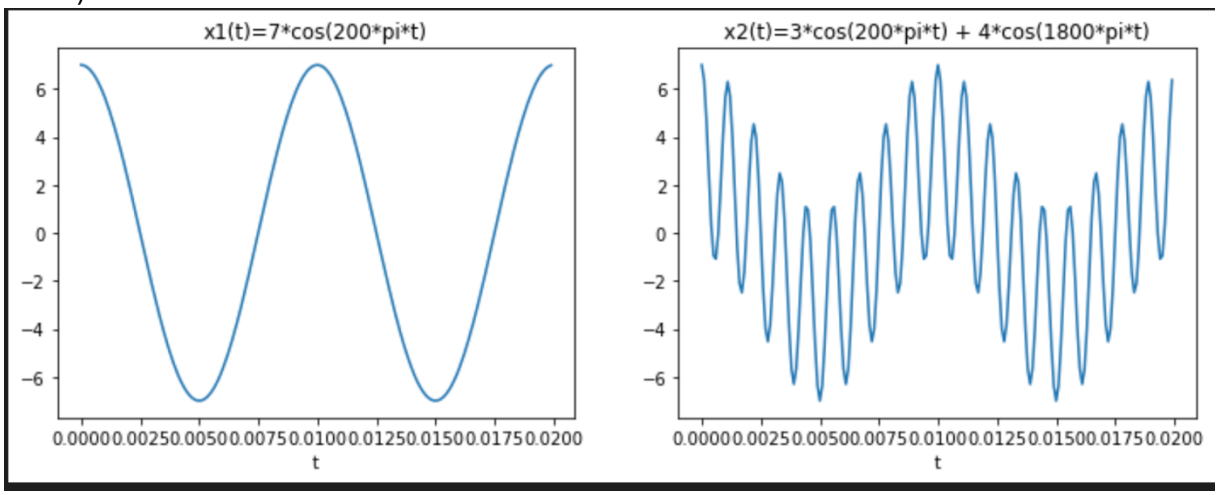
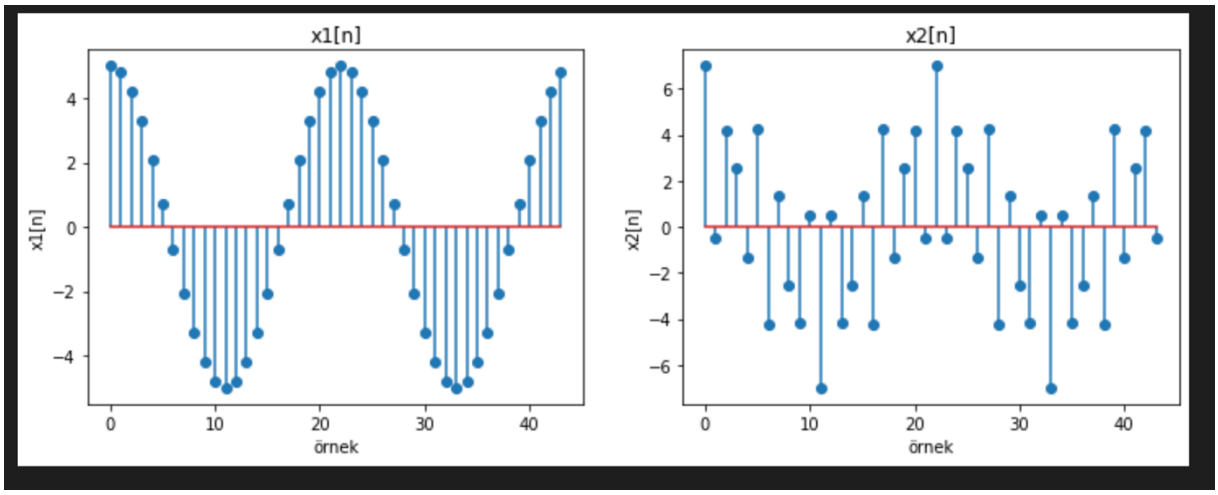
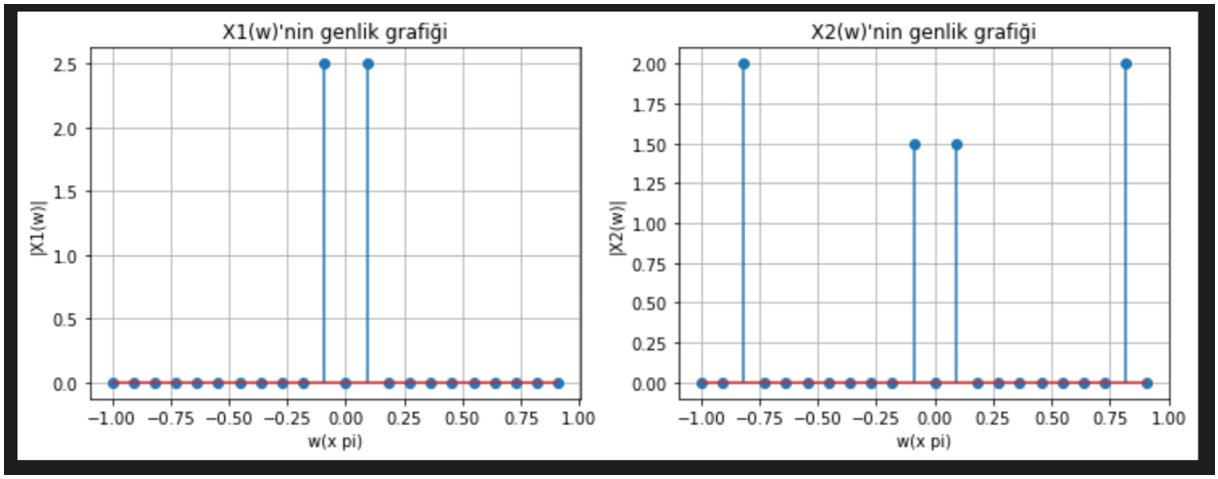
1-a-b)


x1(t)=7*cos(200*pi*t)
x2(t)=3*cos(200*pi*t) + 4*cos(1800*pi*t)
t
t


x1[n]
x2[n]
x1[n]
x2[n]
örnek
örnek


X1(w)'nin genlik grafiği
X2(w)'nin genlik grafiği
|X1(w)|
|X2(w)|
w(x pi)
w(x pi)

```
x1 işaretinin frekans bileşenleri
w =  -0.09090909090909143 X1(w) =  2.5
w =  0.09090909090909029 X1(w) =  2.4999999999999996
x2 işaretinin frekans bileşenleri
w =  -0.8181818181818182 X2(w) =  2.0000000000000013
w =  -0.09090909090909143 X2(w) =  1.4999999999999991
w =  0.09090909090909029 X2(w) =  1.4999999999999993
w =  0.8181818181818171 X2(w) =  2.0000000000000013
```

Birinci işarette -pi/11 ve pi/11 de genliği 2.5 olan dirac görülmektedir bunun ters fourier dönüşümü bu şekildedir ve örneklenmiş işaretle aynıdır $5cos(\pi n/11)$
İknci işaretin -pi/11 ve pi/11 de genliği 1.5 olan dirac görülmektedir ve -9pi/11 ve 9pi/11 de genliği 2 olan dirac görülmektedir. Bunun ters fourier dönüşümü bu şekildedir ve örneklenmiş işaretle aynıdır $3cos(\pi n/11) + 4cos(9\pi n/11)$

2

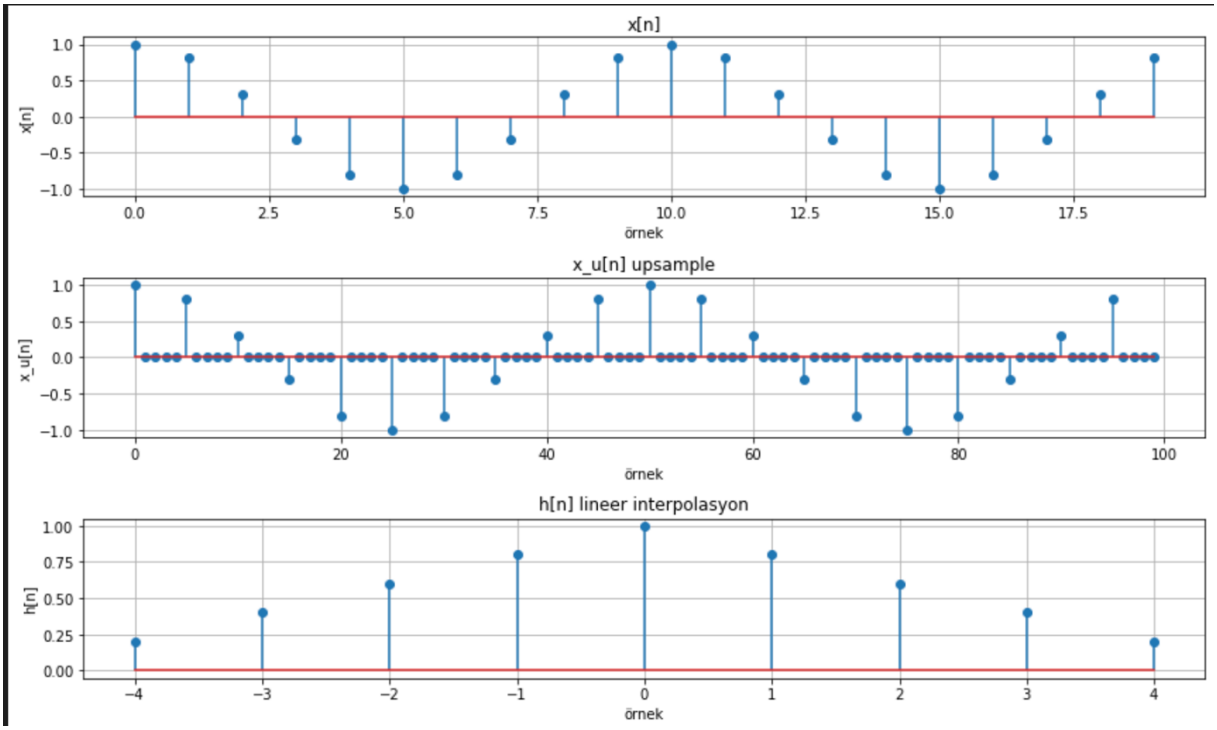
a)

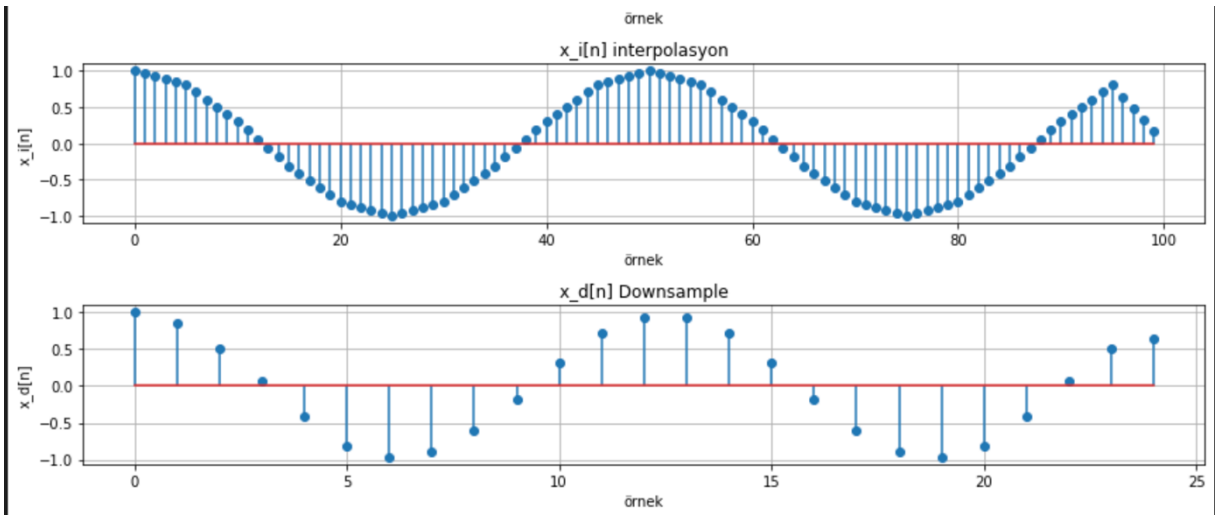
Örnekleme frekansı 40 kHz olan $x[n]$ işaretinin örnekleme frekansının 50 kHz olması için yapılması gereken sık ve seyrek örnekleme oranlarını belirleyiniz ve aşağıdaki soruları belirlediğiniz değerleri kullanarak cevaplayınız.

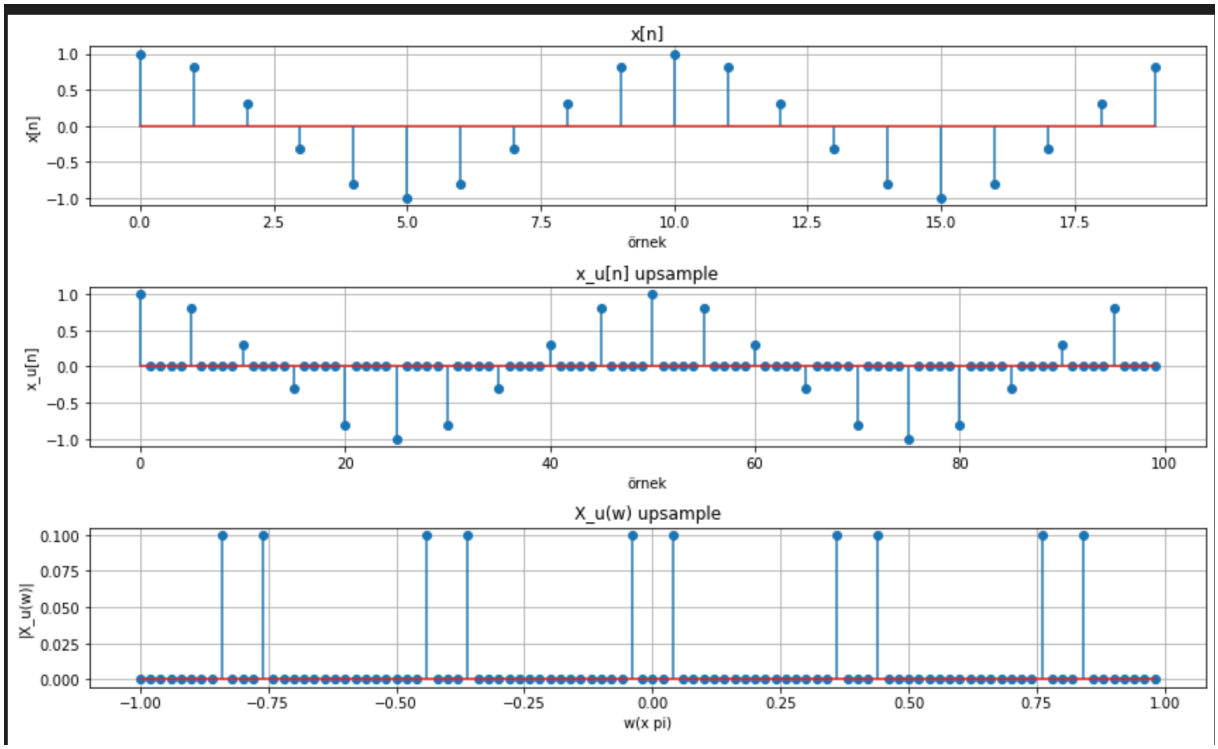
NOT-1: a seçeneğinde L ve M değerlerini bulurken gerekli sadeleştirmeleri yaparak olabilecek en küçük değerleri bulunuz.

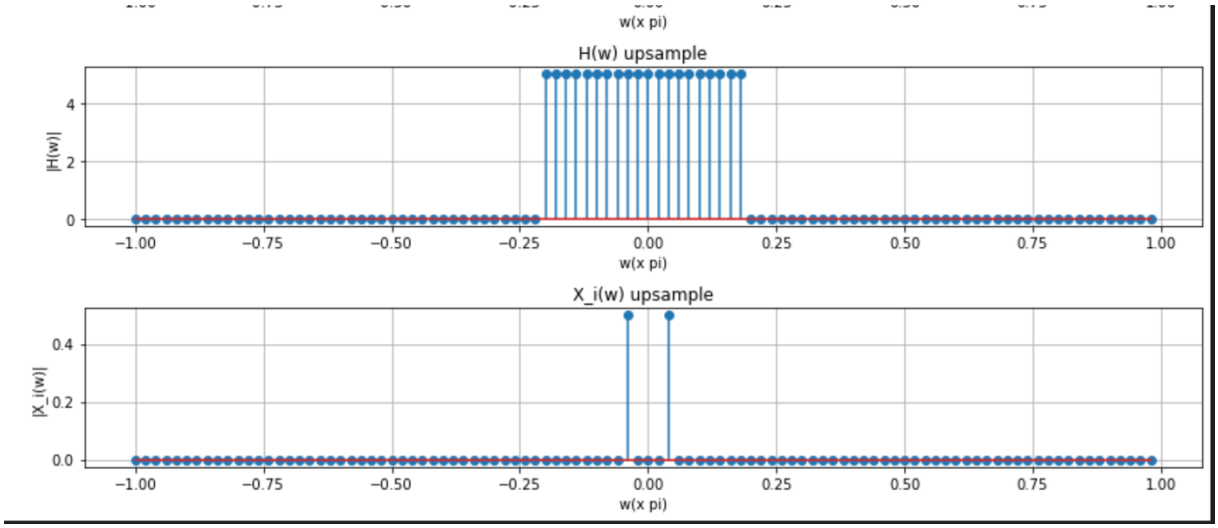
40 *5 = 200 => 200/4 = 40 Khz o zaman
L = 5 = upsample faktörü
M = 4 = Downsample faktörü
olur

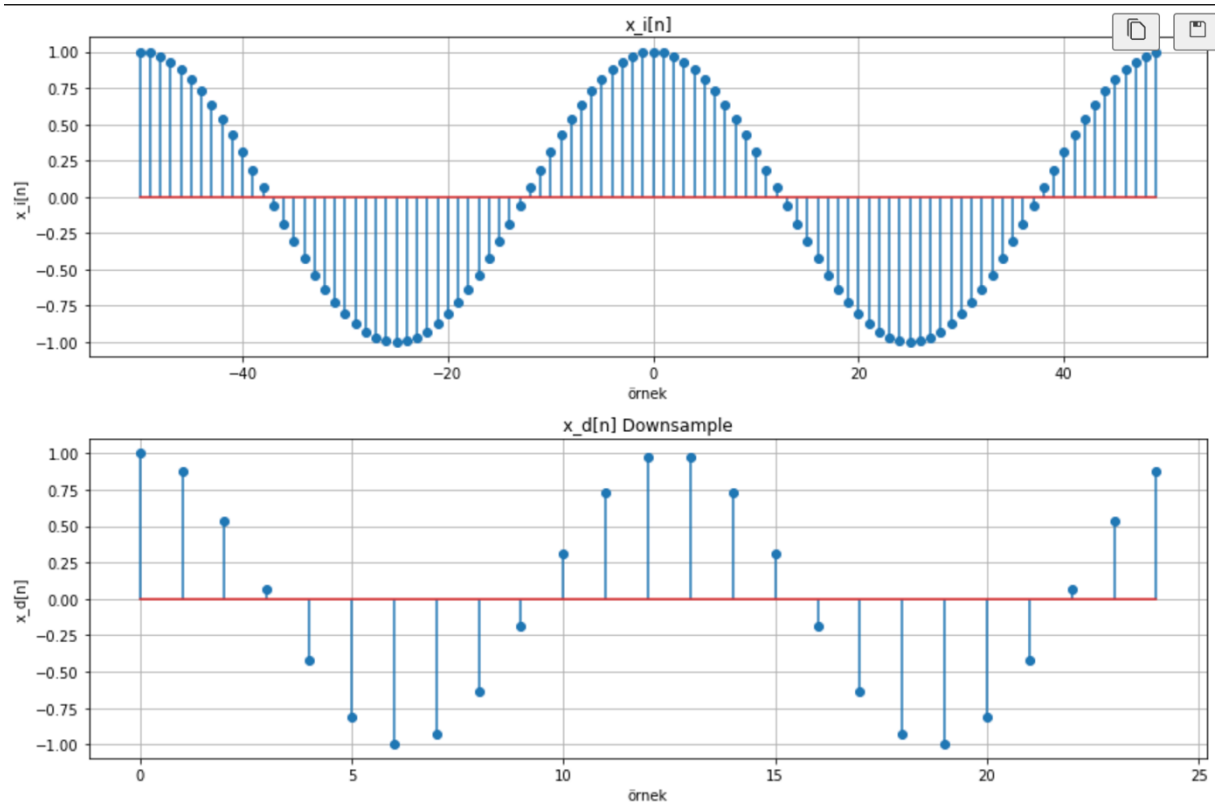
b)




örnek
x_i[n] interpolasyon
x_i[n]
örnek
x_d[n] Downsample
x_d[n]
örnek


x[n]
x[n]
örnek
x_u[n] upsample
x_u[n]
örnek
X_u(w) upsample
|X_u(w)|
w(x pi)


w(x pi)
H(w) upsample
|H(w)|
w(x pi)
X_i(w) upsample
|X_i(w)|
w(x pi)


x_i[n]
x_i[n]
örnek
x_d[n] Downsample
x_d[n]
örnek

3

a)

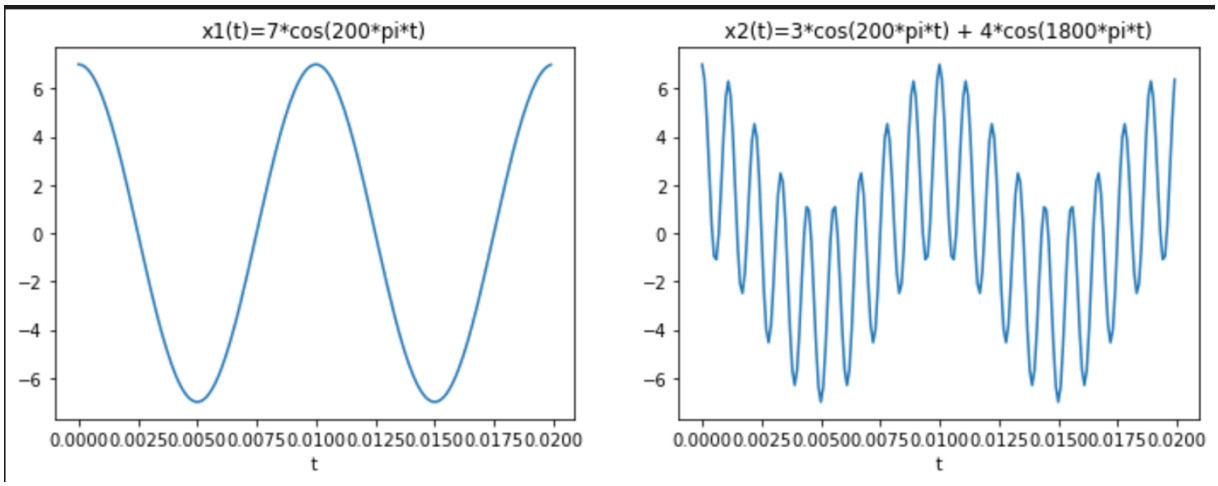
Ödev 5 EK

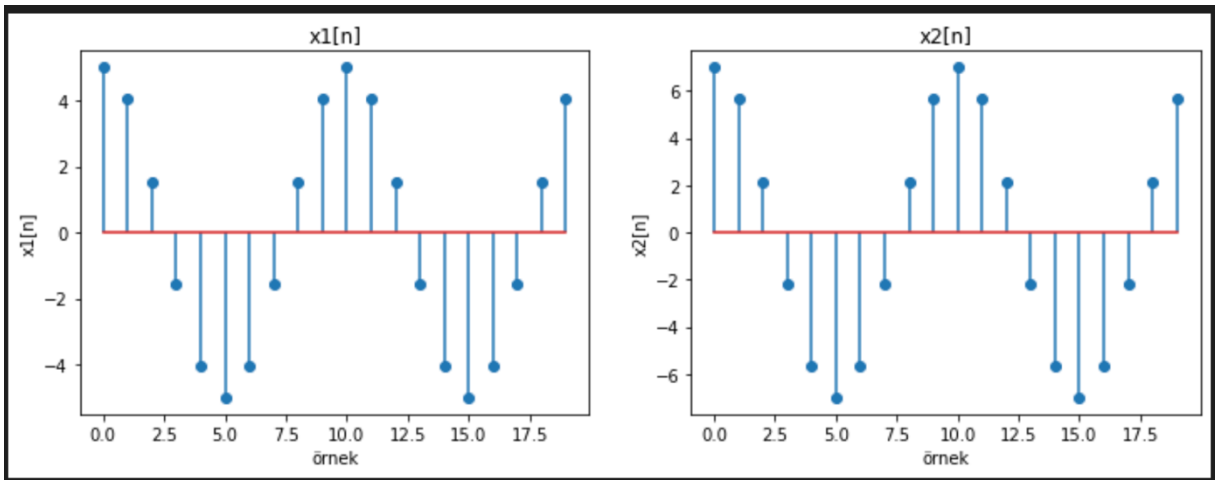
SORU-3: Soru-1'i Fs = 1000Hz için tekrarlayınız.

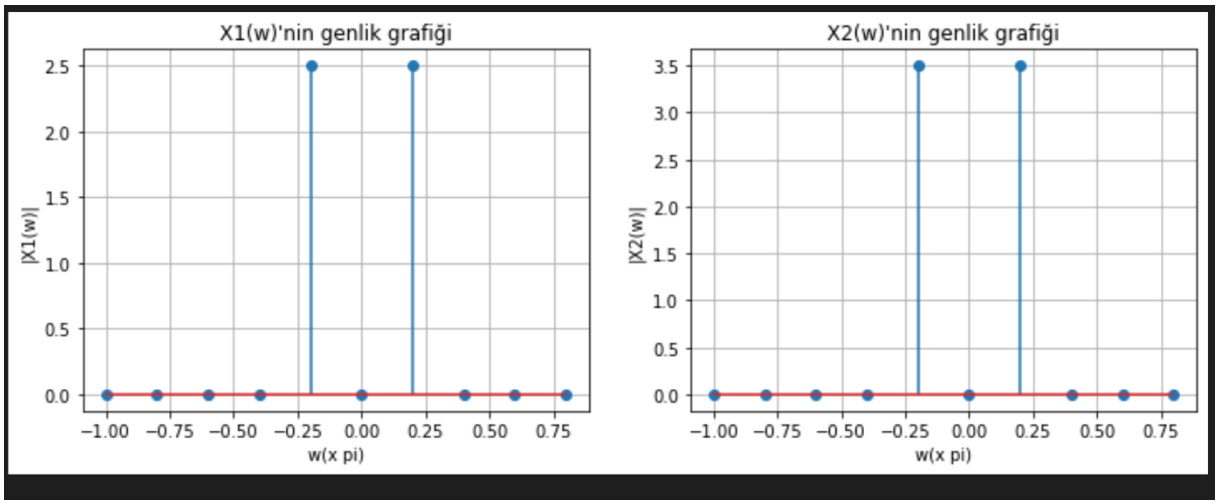
$x1(t)$ = 7cos(200πt) ve $x2(t)$ = 3 cos(200πt) + 4cos(1800πt) işaretlerini Fs = 2200Hz ile örnekleyiniz.

a)

Örnekleme sonucunda elde ettiğiniz $x1[n]$ ve $x2[n]$ sinyallerinin grafiklerini iki periyot için çizdirerek karşılaştırınız. İki işaret arasında her hangi bir fark var mı? Neden?



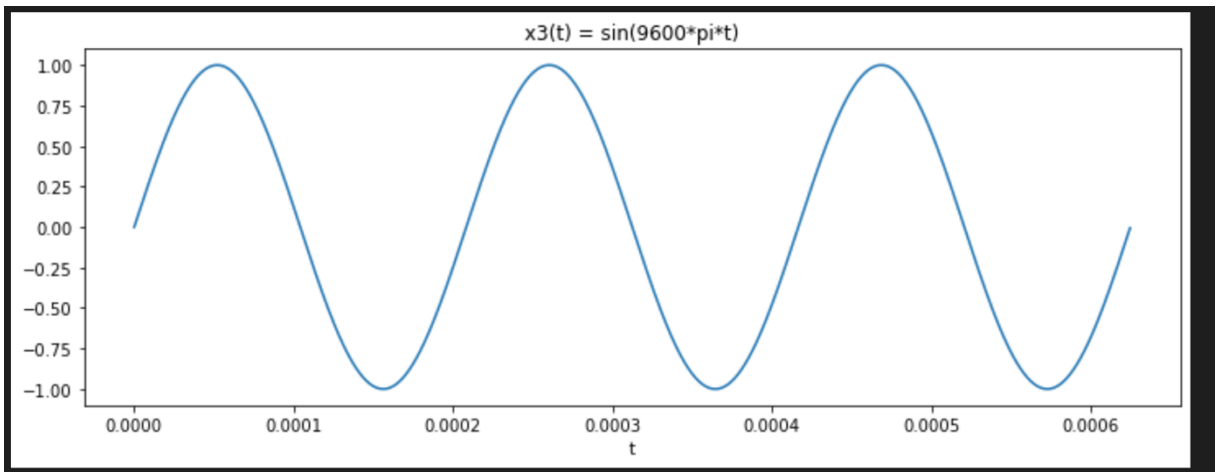


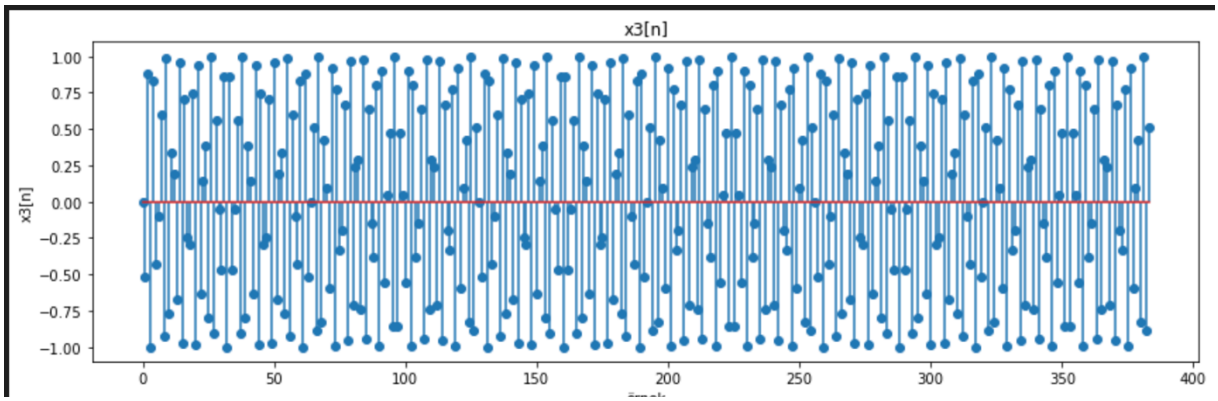


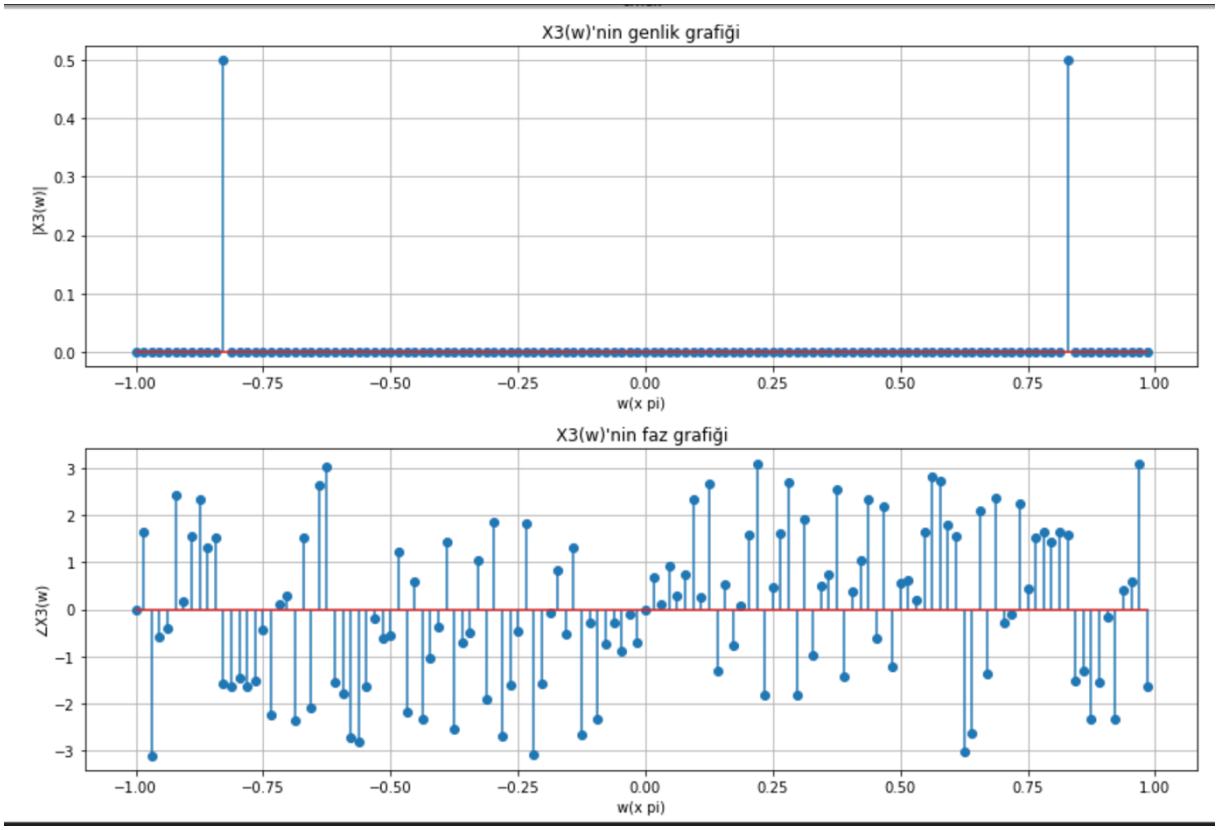
3c) wları pi ile çarpılması lazım yanında pi yazmıyorsa da aslında pi var yaparken untmuş olabilirim

```
x1 işaretinin frekans bileşenleri
w =  -0.2 X1(w) =  2.5
w =  0.2 X1(w) =  2.5
x2 işaretinin frekans bileşenleri
w =  -0.2 pi X2(w) =  3.500000000000002
w =  0.2 pi X2(w) =  3.500000000000002
```

4 –
N = 128 olarak bulundu !!!



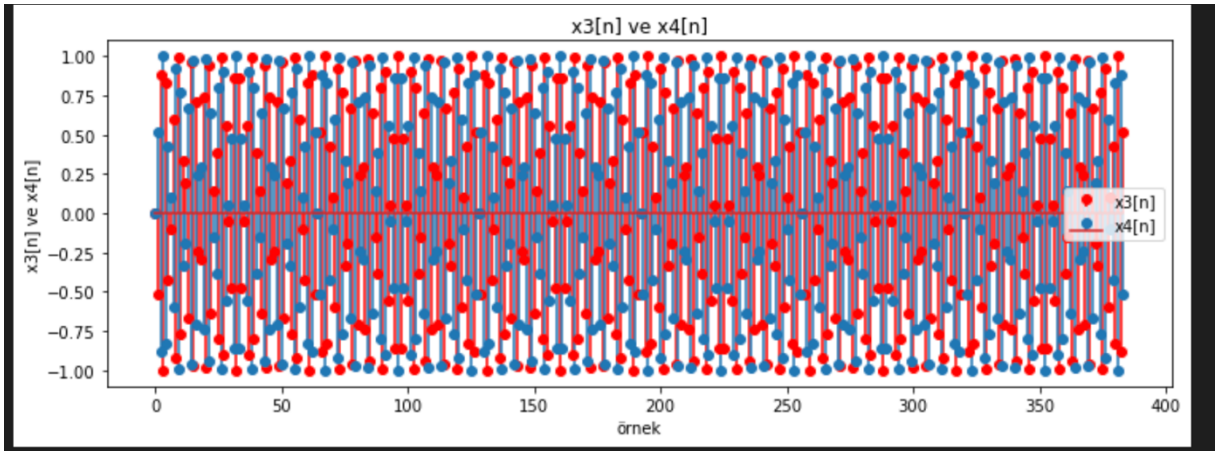




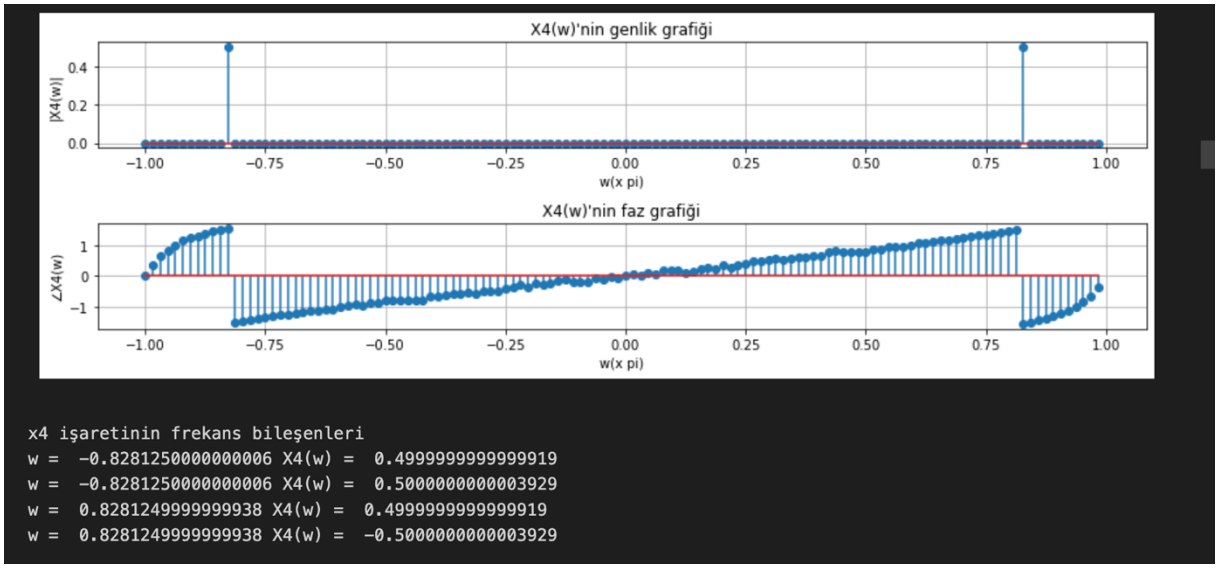
4d)

```
x3 işaretinin frekans bileşenleri
w =  -0.8281250000000006 X3(w) =  0.49999999999999994
w =  -0.8281250000000006 X3(w) =  -0.5000000000000038
w =  0.8281249999999938 X3(w) =  0.49999999999999994
w =  0.8281249999999938 X3(w) =  0.5000000000000038
```

Wlar pi ile çarpılmalı



Aynı çıkmadılar !!!!!!!!



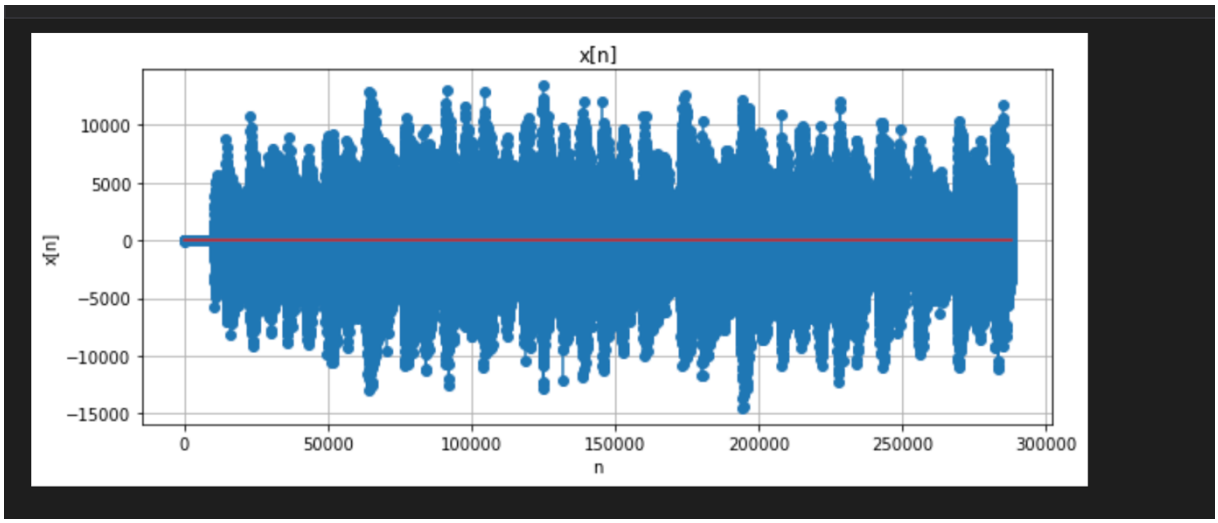

```
x4 işaretinin frekans bileşenleri
w =  -0.8281250000000006 X4(w) =  0.4999999999999919
w =  -0.8281250000000006 X4(w) =  0.5000000000003929
w =  0.8281249999999938 X4(w) =  0.4999999999999919
w =  0.8281249999999938 X4(w) =  -0.5000000000003929
```

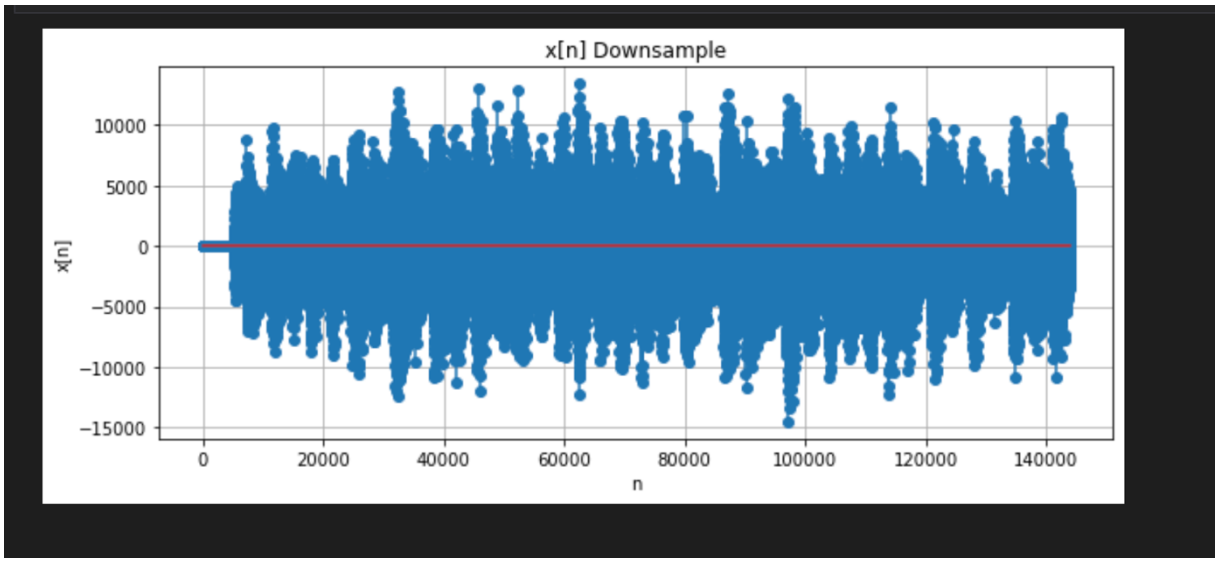
Ama fourierleri aynı labta karşılaştığımız sorunun aynısı !!!!

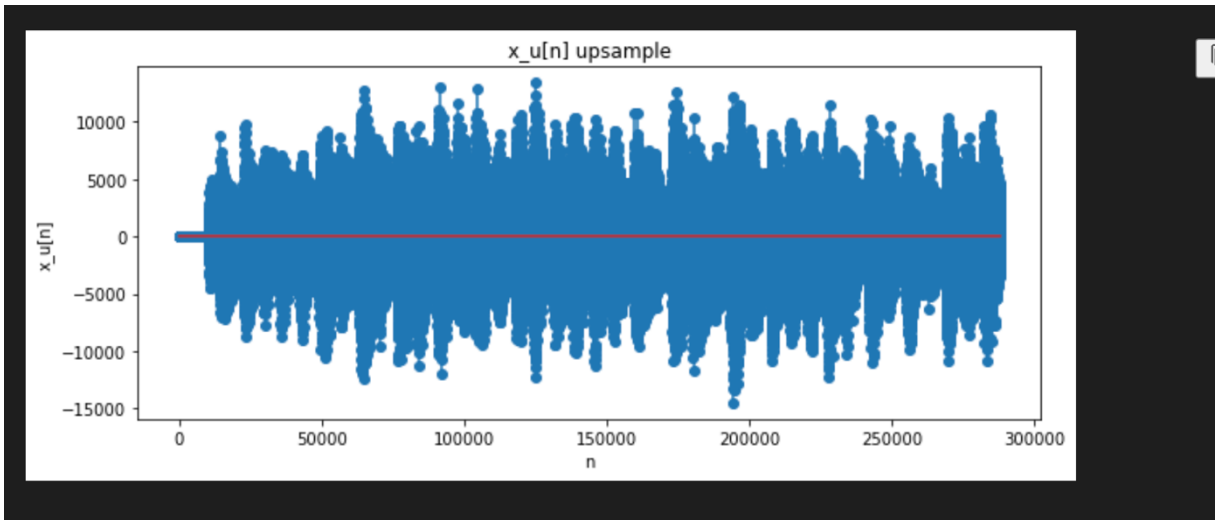
5a)

```
print(x.shape) # ses dosyasının boyutunun yazdırılması
print(Fs) # ses dosyasının örnekleme frekansının yazdırılması
```

```
(288000, 2)
48000
```

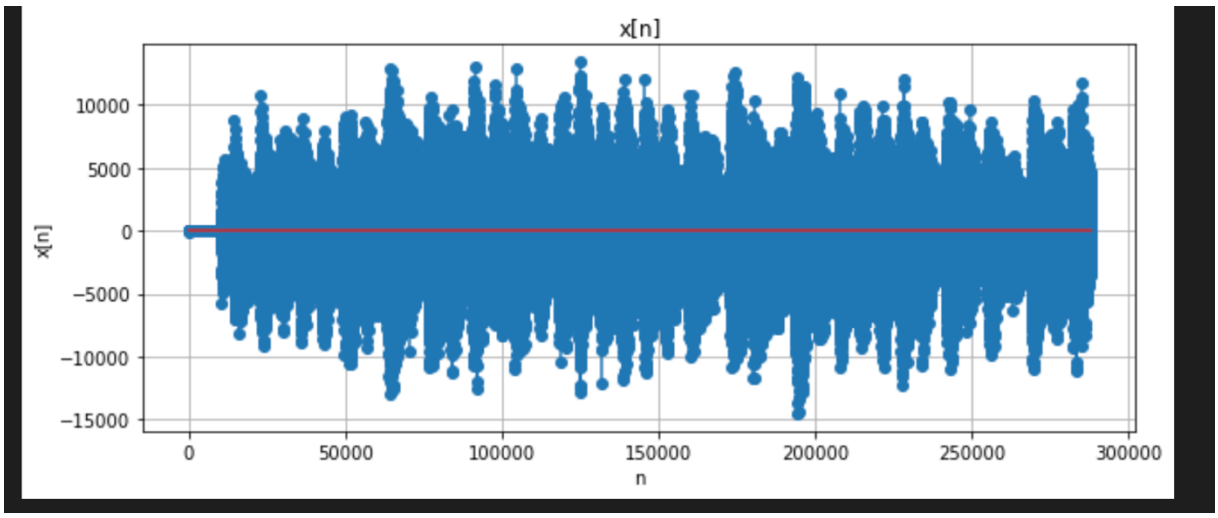


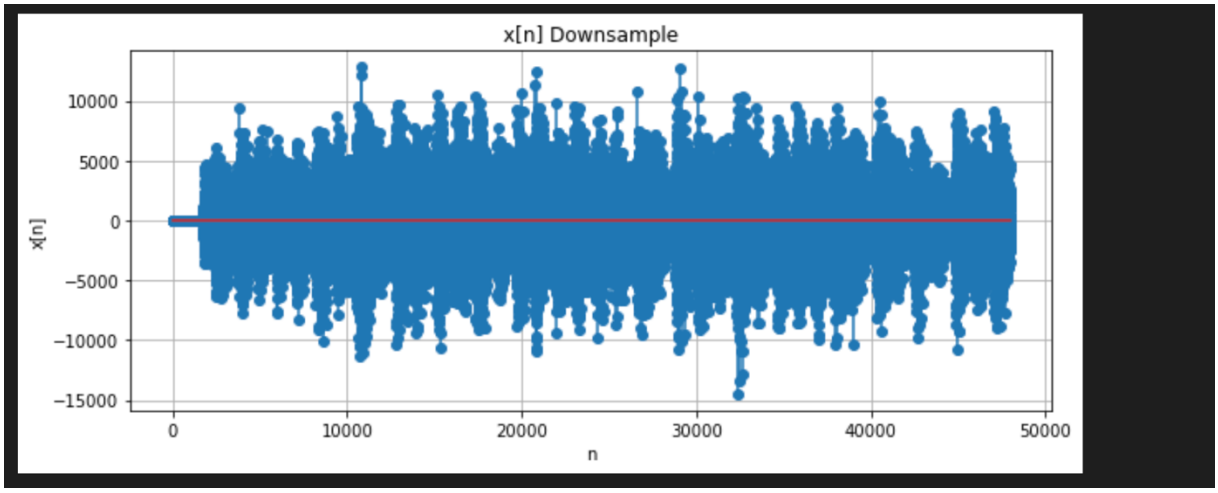


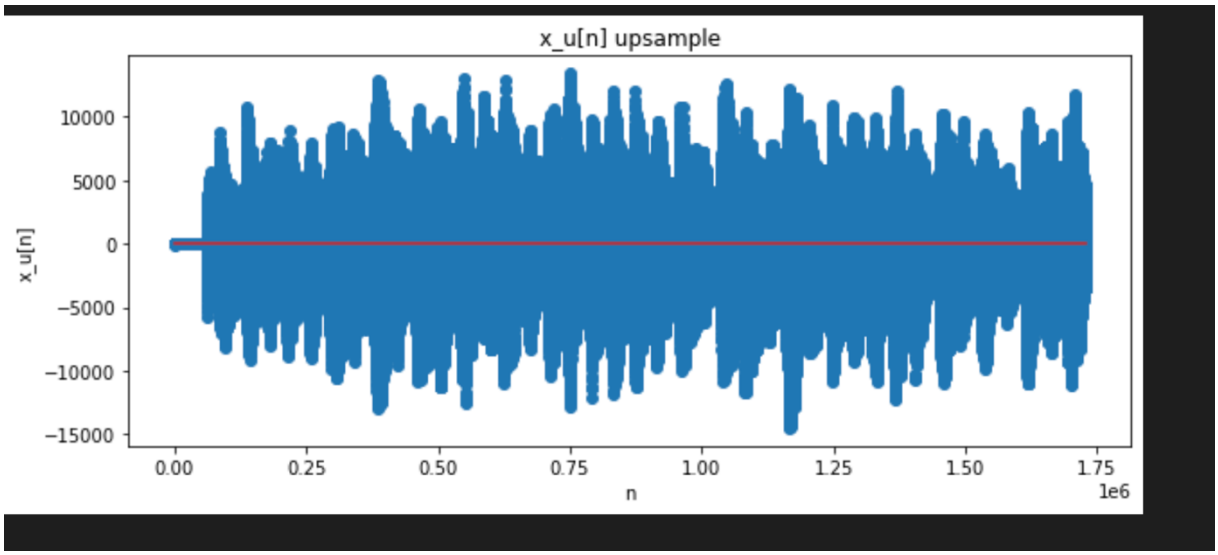


FFT İLE YAOILANIN GRAFİKLERİ GEREKTİRDİĞİ YÜKSEK İŞLEM GÜCÜ VE ZAMAN VE DEPOLAMADAN DOLAYI ÇİZDİRLMEMESİ TERCİH EDİLMİŞTİR. SABRIMIZ AZ KARDEŞ...

b)

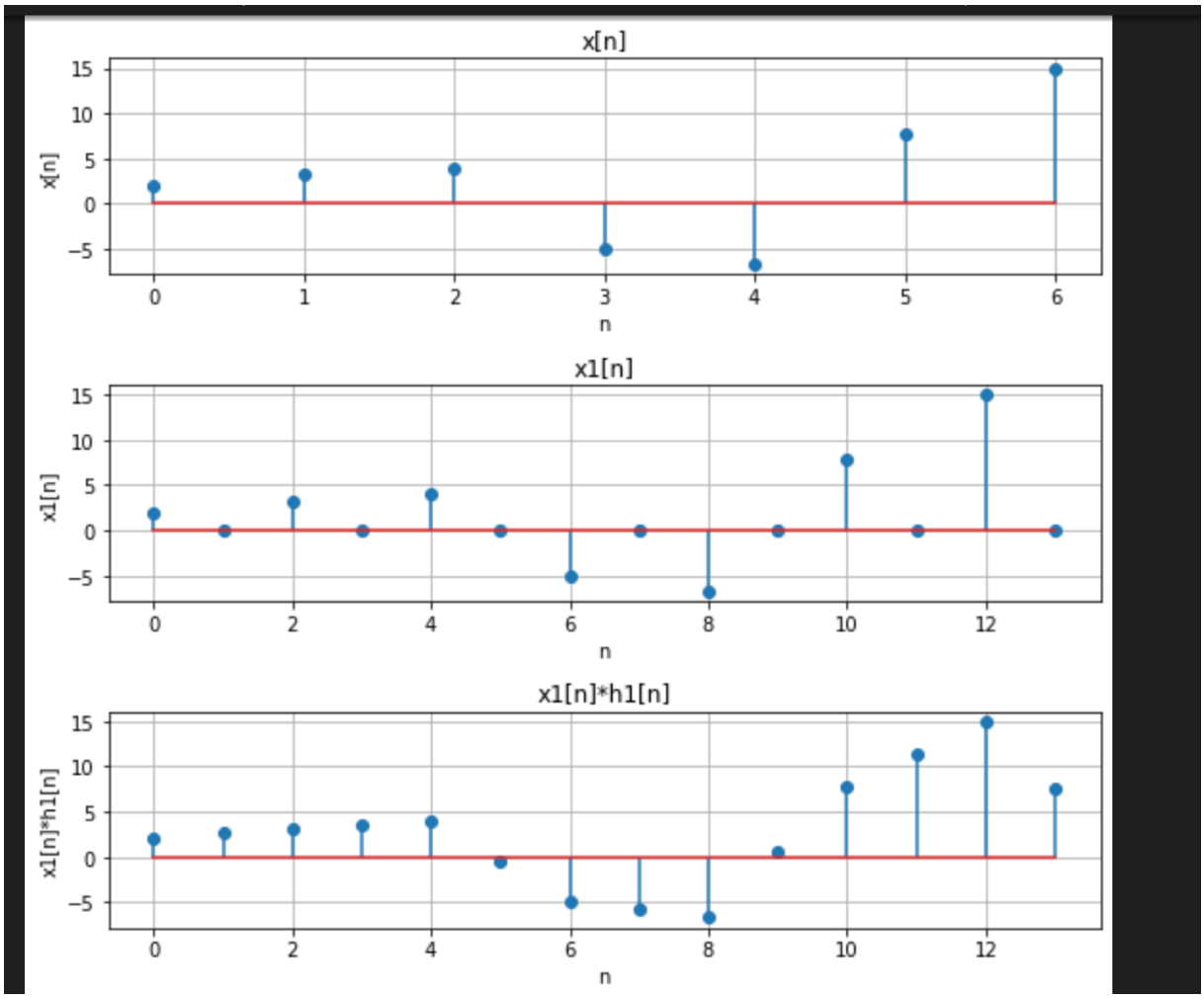

x[n]
10000
5000
0
−5000
−10000
−15000
x[n]
0
50000
100000
150000
200000
250000
300000
n

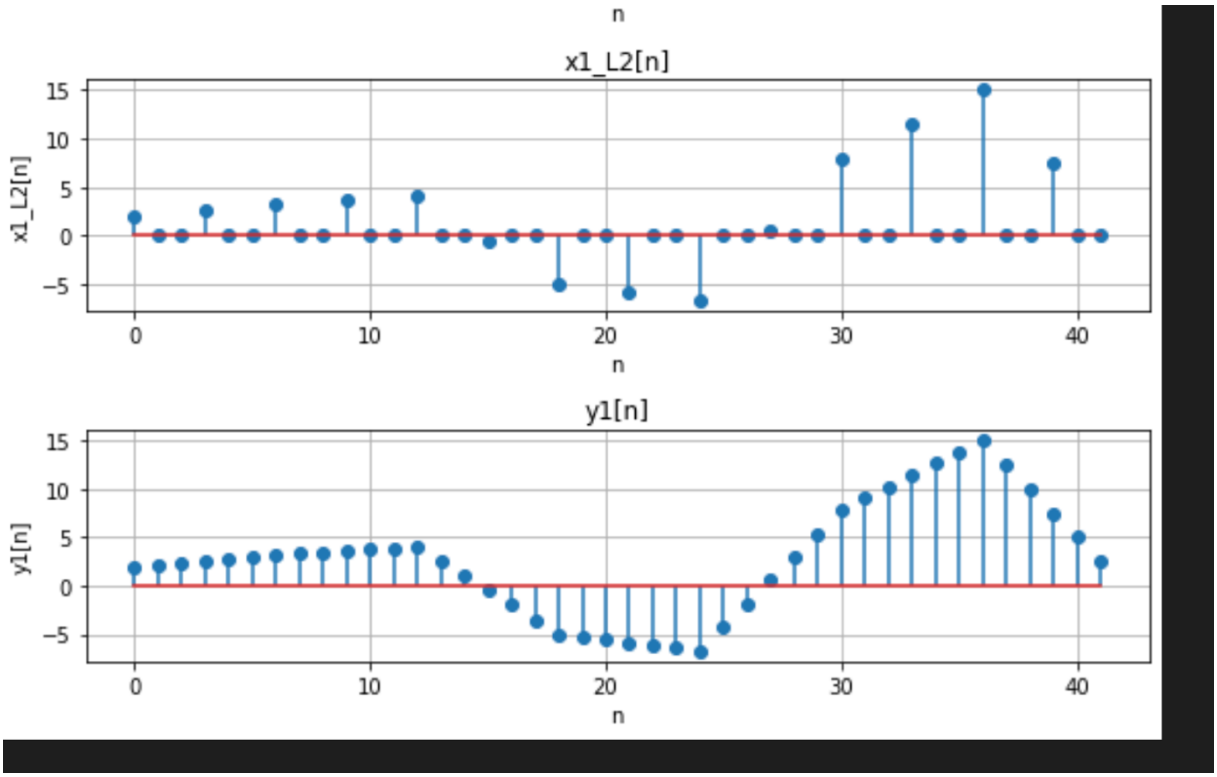

x[n] Downsample
10000
5000
0
−5000
−10000
−15000
x[n]
0
10000
20000
30000
40000
50000
n

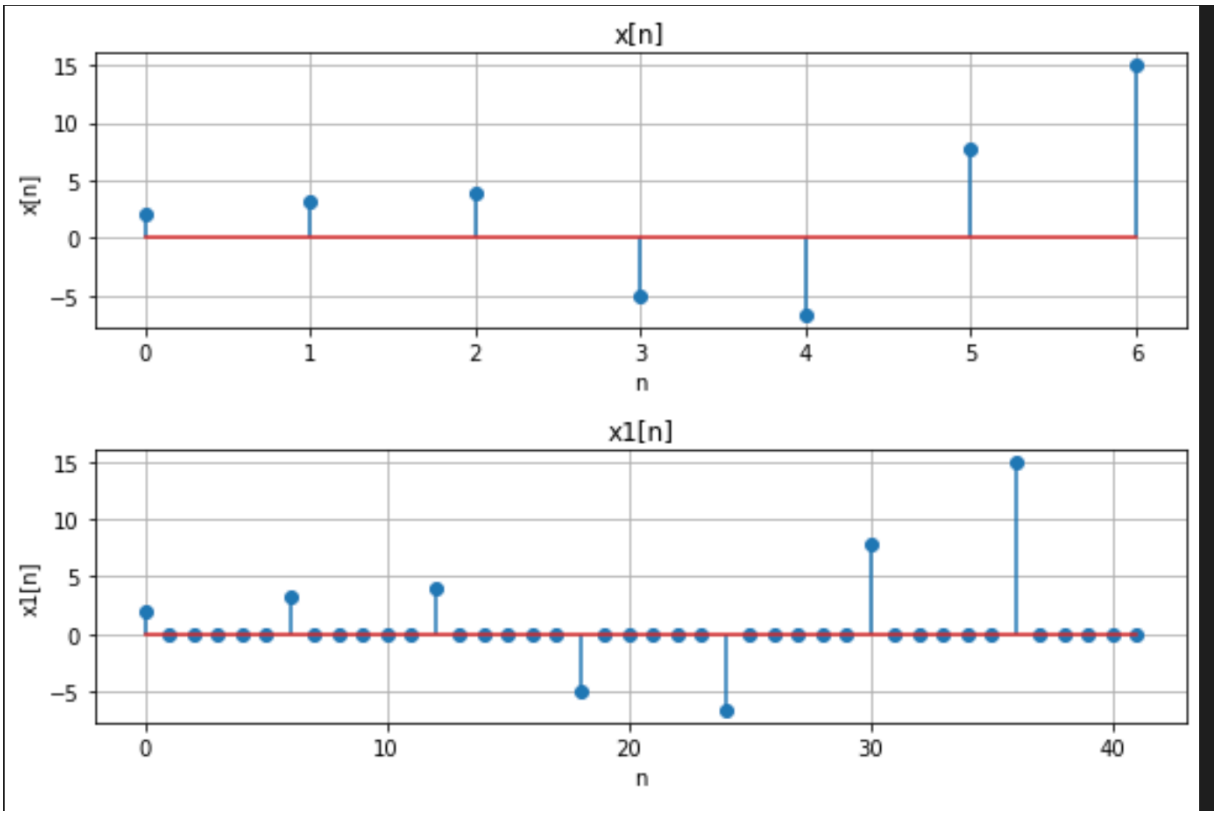


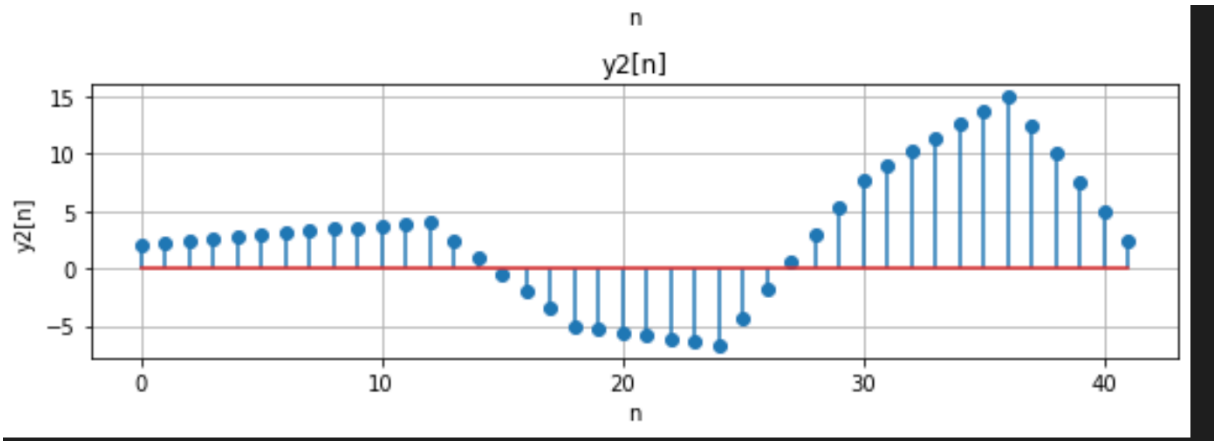
Upsample fft ile çizdirilmemiştir.

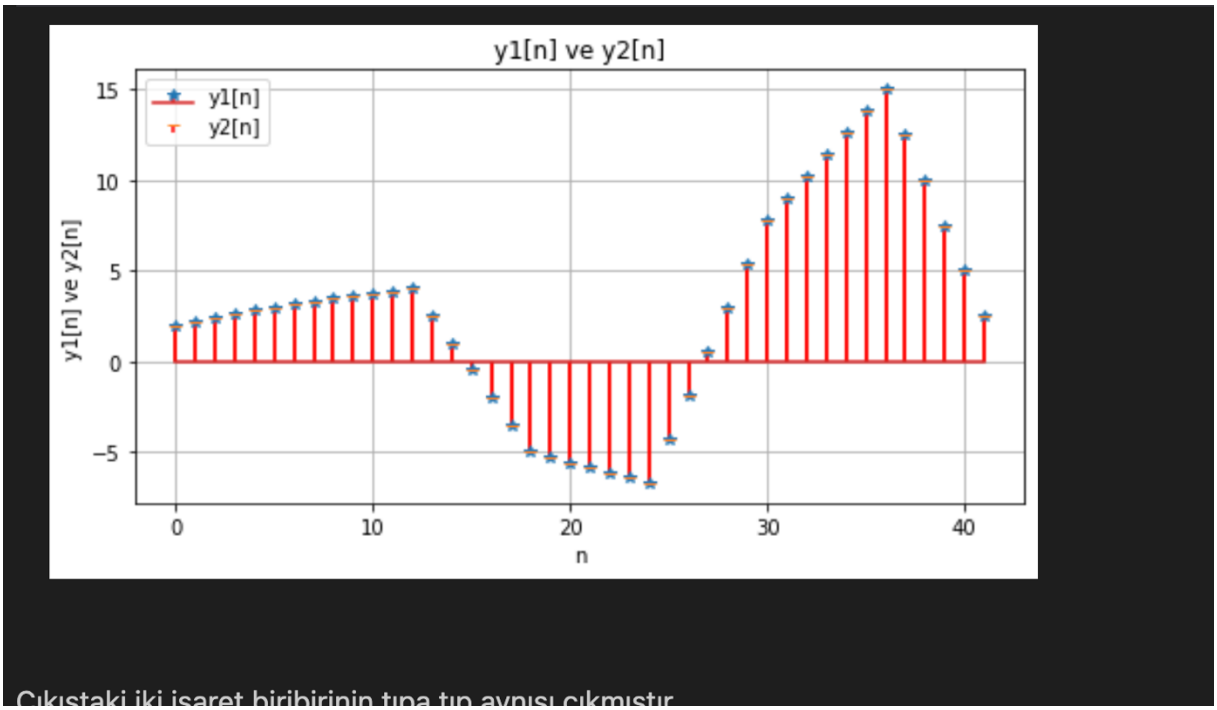
6a)




n
x1_L2[n]
x1_L2[n]
n
y1[n]
y1[n]
n


x[n]
x[n]
n
x1[n]
x1[n]
n





Eh bitti işte...